Competency	Competency Level	Contents	Learning outcomes	Periods
	13.3 Analyzes the existing models of computing and proposes new models	• Beyond von-Neumann computer • Nature inspired computing • Biology inspired computing • Fundamentals of quantum computing • Applications	• Predicts the technologies beyond von-Neumann computers	4

Competency 14:
Designs and Implements a simple Information system project
	14.1 Conducts a project on designing an information system	• Examples of projects
		• Stakeholders
		  ○ Roles and responsibilities of the following stakeholders:
		  • senior management
		  • customer/client
		  • user
		  • project manager
		  • team member
		  • peer reviewer
		  • supplier
		• Project planning
		  ○ The phases of the project
		  ○ The activities to be carried out in each phase
		  ○ Start date and end date of each activity
		  ○ Dependencies
		  ○ Resources required for each activity
		  ○ Dates of key milestones
		  ○ Potential risks, their effect on the plan and how their impact can be minimized
			• Identifies the characteristics and reasons of projects with examples
			• Identifies the role of stakeholders
			• Identifies the reasons of project plan and the planning criteria
			• Identifies a simple problem to be carryout as a project
			• Prepares project proposal
			• Presents the proposal
			• Organizes the project
			• Carryout the project according to the SDLC
				25
				
				One period per week for a duration of one year

Competency	Competency Level	Contents	Learning outcomes	Periods
		○ Gantt charts • Identification of a simple problem for the project • Project proposal ○ Proposal preparation ○ Getting approval • Project organization ○ Storing documents relating to a project (project folder) ○ Protecting information from accidental damage ○ Communicating with stakeholders ○ Reporting on progress ○ Holding reviews • Phases of a project ○ Preliminary investigation ○ Feasibility study ○ Requirement analysis ○ Design ○ Coding ○ testing ○ Documentation	phases • Handovers the outcomes of each phase to documentation after evaluate by the teacher	
	14.2 Implements and demonstrates the Information system	• Project presentation and demonstration of the system	• Presents the finished project and demonstrates the system to all the class	05

10.0 Learning -Teaching Process

Information and Communication Technology is not only a dynamic subject; it is also a living subject. It is necessary to update one's knowledge in order to be alert to the latest findings in this area. ICT has taken such an important position that the percentage of literacy in ICT has come to be regarded as an index of development. It is essential that the proposed method of teaching should be student-centered as this subject is essentially practice-oriented. In view of the fact that this is an essentially practical subject, it is vital that a student is involved in self study apart from the day to day learning- teaching process in order to ensure that subject matter learned is firmly grasped. There is need for special attention to be paid to the encouragement of the student for self-study. Further, just as much as knowledge, attitudes and skills, as well as social values, legal constraints related to the subject carry with them the imperatives of a sense and spirit of self-discipline. It is essential that the learning-teaching evaluation process is so organized as to highlight the importance of computer use.

The global trend in present day education is to introduce competency-based curricula, which promote collaborative learning through student centered activities where learning predominates teaching. It is intended for the students to actively participate in activities, which enhance the development of individual social and mental skills. The following aspects are emphasized.

1. **Teachers have the freedom to follow most suitable teaching method to achieve the desired competencies.**
2. **Allow the students to acquire hands on experience through self-directed activities.**
3. **Direct students to acquire knowledge and information through reliable sources wherever necessary.**

Competency 8: Designs and develops database systems to manage data efficiently and effectively

Competency Level 8.1: Learns the basics of information and data, and the need for databases

Time: 02 periods

Learning Outcomes:

- Distinguishes data and information
- Defines database
- Lists and briefly describes the database models
- Compares and contrasts database models in terms of their features

Contents:

- Data vs. information
- Structured Vs. unstructured data
- Definition of database
- Database models
  - Flat file system
  - Hierarchical model
  - Network model
  - Relational model
  - Object relational model
- Comparison of database models

Concepts and terms to be highlighted:

- Data
- Information
- Need for transformation data (unstructured) into information (structured)
- Categorization of data into group(tables)
- Advantage of databases and there feathers.
- Deferent types of database models
  - Database models
    - Flat file system
    - Hierarchical model
    - Network model
    - Relational model
    - Object relational model
  - Comparison of database

Guidance for lesson plans:

- Divide the students into groups.
- Ask groups to collect data from particular instance.
  - Ex: Select twenty books randomly selected in school library find the detail information of each books (Title, Publication date, ISBN No, Author, etc.)

- Ask the groups to put the collect data into meaningful group

    - Ex: group the books name according to subject
- Discuss with the student which data or information is more useful.

- Discuss the organizing of data in some order and how efficient to search for an item in a sequence
  Ex: 1. according to alphabetical order in author's name
  2. Search for an item when there in order (author name or subject name).
- Discuss the importance of databases
- Discuss t advantages and features of database
- Discuss the deferent database models
- Compare the different database models

Guidance for assessments and evaluations:

- Divide students into group and ask them to find real world examples and identify data and information and categories into a suitable tables.
- Ask the group to present their findings
- Ask student to identify witch category their example belong to
- Ask the student group to prepare a presentation about the advantages and features of database

Quality inputs:

Library access facility

Presentation

Competency Level 8.2: Describes the main components of the relational database model

Time: 04 periods

Learning Outcomes:

- Defines relations / tables
- Names and describes main components of a relational database
- Describes the relationships in terms of relational database model
- Briefly explains the types of constrains

Contents:

- Relations / Tables
  - Attributes / Columns
  - Tuples / Rows
  - Relationships
- Types of Constraints
  - A NOT NULL Constraint
  - A Unique Constraint
  - A Primary Key Constraint
  - A Foreign Key Constraint
  - A (Table) Check Constraint

Concepts and terms to be highlighted:

- Relations / Tables
  - Attributes / Columns
  - Tuples / Rows
  - Relationships
- Types of Constraints
  - A NOT NULL Constraint
  - A Unique Constraint
  - A Primary Key Constraint
  - A Foreign Key Constraint
  - A (Table) Check Constraint

Guidance for lesson plans:

- Provide a print out of created tables with relationships and discuss with the student the features (table, attribute, tuples, and relationships) of them.
- Divide students into groups
- Provide print outs with improperly created tables and well-designed tables.
- Compare and discuss the above table.
- Request to group to identify the inappropriate design of tables.

- Discuss the needs for constraints for the design
- Explain the type of constraints for the design.

Guidance for assessments and evaluations:

- Provide printouts of created tables and relational schemas to students and ask them to identify the attributes, tuples, relationships and the applied constraints.
- Provide printouts of created tables and relational schemas to students and ask them to identify their shortcoming.

Quality inputs:

- Printouts of created tables and relational schemas, presentation

Competency Level 8.3: Analyzes the main components of a database system

Time: 14 periods

Learning Outcomes:

- Lists and briefly describes the component of a database system
- Describes the database management system
- Defines SQL
- Distinguishes between DDL vs. DML
- Uses appropriate SQL commands for creating and using database
- Uses appropriate commands to create tables with suitable fields and data types
- Sets primary key and foreign key while creating table
- Uses primary key and foreign key after completion of a table
- Creates relationships among tables
- Uses appropriate SQL commands to Insert and delete columns, delete foreign key / primary key and to drop table
- Uses appropriate SQL commands to drop database
- Uses appropriate commands to Insert, modify retrieve, update and delete data.
  Uses appropriate DML commands to query data according to the requirements

Contents:

- Data Base Management System
- Data definition language (DDL)
  - Introduction to SQL
  - Classification of SQL
  - Creating, using relational database using DDL
    - ❖ Creating table
    - ❖ Alter table
      - Inserting and deleting attributes
      - Adding and deleting foreign key and primary key
    - ❖ Drop tables
    - ❖ Drop databases

- Data manipulation Language (DML)
  - DML Features in SQL
    - ❖ Inserting, modifying, retrieving, updating deleting data
  - Select Query
    - ❖ Extracting rows and columns from single table
    - ❖ Extracting rows and columns from multiple tables using inner join operation
    - ❖ Insert Query
    - ❖ Update Query
    - ❖ Delete Query

Concepts and terms to be highlighted:

- Database management systems
- Understand how databases are implemented and used in practice
- Selection and use of Primary Key and Foreign Key
- DDL (Data Definition Language) : The language used to create the database structure

- The functions of DDL commands – Examples - Create, Drop, Alter, etc.
- DML (Data Manipulation Language): The language used to manipulate data
- The functions of DML Commands: Examples – Insert, modify, delete et
- Database Query: Standard commands used to query a database according to given conditions. Example – Finding the names of students living in a particular town

Guidance for lesson plans:

- Introduce the concept of DBMS
- Discuss the functions of DBMS with examples
- Distribute a practical sheet of command (DDL and DML) summary and request students to practice the commands
- Discuss the criterion for selecting smelting and appropriate the of keys (primary and foreign)
- Discuss some sample queries for a given condition
- Provide some conditions and ask students to make appropriate queries satisfying the conditions

Guidance for assessments and evaluations:

- Provide the students with a worksheet to create a DBMS and to use it.
- This worksheet should contain all the commands students learnt. In addition, it should contain few logical queries that could be done on the DBMS.

Quality inputs:

- Computers with DBMS software, Practical sheets, Worksheets, Presentations

Competency Level 8.4: Designs the conceptual schema of a database

Time: 12 periods

Learning Outcomes:

- Describes ER diagram
- Describes the components of an ER diagram (entities, attributes)
- Describes entity identifiers
- Lists and describes relationships
- Describes cardinality
- Identifies the requirements of a given scenario
- Selects entities, attributes and according to the requirement
- Draws the ER diagram
- Explains the EER diagrams

Contents:

- ER (Entity Relationship) diagrams
  - Entities, attributes
  - Entity identifiers
  - Relationships
  - Cardinality

  Introduction to EER (Extended ER) diagrams

Concepts and terms to be highlighted:

- Decomposes a real world systems into logical subsystems (relations) and to connect (relationship) them logically

- ER (Entity Relationship) diagrams
  - Entities
  - Attributes
  - Relationships
  - Cardinality
- Introduction to EER (Extended ER) diagrams

Guidance for lesson plans:

- Divide students into groups and provide a scenario to each group
- Ask the group to identify the nouns and verbs in the provided scenario
- Request the group to identify the nouns which are independent and dependent
- Request the group to appreciate how independent nouns become Entities and dependent nouns become the description (attributes) of the independent nouns.
- Request group to appreciate how the verb becomes a connections (relationship) between two independent nouns(Entities)
- Request group to identify the connection between entities and identify the multiple number of connections between the attribute of two entities.

- Discuss that witch connection are called the cardinality (one to one, one to many, many to many)
- Request student to identify why ER diagram for not suitable for specialization and generalization of entity.
- Discuss Extended Entity Relationship (EER) diagram.
- Construct and discuss some examples of ER diagram and EER diagrams.

Guidance for assessments and evaluations:

- Request student to identify the entities and relationships for a given scenario.
- Request student to construct ER or EER diagram.
- Request student to identify the cardinality ratios and illustrate it properly in to constructed diagram.

Quality inputs:

Scenarios, illustration from web and internet sources (ex: YouTube)

Competency Level 8.5: Designs the logical schema of a database
Competency Level 8.6: Transforms ER diagrams to logical schema

Time: 12 periods

Learning Outcomes:

- Defines logical schema of a database
- Describes relational schema
- Describes relational instances
- Briefly describes Candidate key, primary key, alternate key and foreign key
- Describes the methods of transformation ER diagram to logical schema
- Transforms ER diagrams (entity, attribute, relationships) to logical schema

Contents:

- Definition of the logical schema
- Database schema design
  - Relational schema
  - Relation instances
  - Candidate key
  - Primary key
  - Alternate key
  - Foreign key
- Domain
- Entity transformation
- Attribute transformation
- Relationship transformation

Concepts and terms to be highlighted:

- Logical schema - tabular structure of representing ER diagram in order to visualize the entities, relationships and relationship keys in a more design oriented way.
- Database schema design tabular method of representing a scenario with relations, relationships and keys.
- Relational schema is another way of drawing the schema diagram.
- Relation instances are the values of the attributes at a given instances.
- Keys are the specific attribute of entities that connect the attributes.
- Keys
  - Candidate key
  - Primary key
  - Alternate key
  - Foreign key
- Domain is the allowed values (types, values, length) of an attribute
  - Ex: in the calendar year 12 months are allowed.

Guidance for lesson plans:

- Divided the student into group
- Guide the groups to convert the construct to ER diagram into relational schema.

- Explain each types keys in the attributes and discuss the appropriate usage of them
- Guide the student to identify and mark the primary key and foreign key in relational schema
- Guide to identify domain of attributes.

Guidance for assessments and evaluations:

- Ask the student to Construct ER diagram for a given scenario
- Ask to student to convert to constructed ER diagram to relational schema and marks the primary and foreign key appropriately.

Quality inputs:

Scenarios, presentation with ER diagram in class, suitable web sites and web facilities. (Ex: YouTube)

Competency Level 8.7: Normalizes database schema to improve performance

Time: 06 periods

Learning Outcomes:

- Describes the functional dependencies and categorizes them
- Describes abnormalities of an improperly designed table when modifying in terms of insert, update and delete
- Describes the zero normal form
- Explains the abnormalities which are reduced after the first normal form
- Lists the conditions for executing the second normal form
- Explains the abnormalities which are reduced after the second normal form
  Explains the abnormalities which are reduced after the third normal form

Contents:

- Need for normalization-
  - Redundancies and anomalies
    - ❖ Insert
    - ❖ Update
    - ❖ Delete
- Functional dependencies
  - Full dependency
  - Partial dependency
  - Transitive dependency
- Levels of normalization
  - Zero normal form
  - First normal form
  - Second normal form
  - Third normal form

Concepts and terms to be highlighted:

- Normalization – improves the performance of a database in inserting, updating and deleting data by reducing redundancies and anomalies.
- Redundancy, Anomaly
- Functional Dependency (FD) – FD is a constraint that describes the relationship among attributes in a table
- Levels of Normalization
  - Zero Normal Form
  - First Normal Form
  - Second Normal Form
  - Third Normal Form

Guidance for lesson plans:

- Divide the students into groups
- Provide the students with a zero normalized tables and ask the groups to identify the shortcomings
- Discuss the identified shortcomings and introduce the important terms like redundancy, anomaly and dependency
- Discuss how redundancy, anomaly and dependency can be reduced using normalization.
- Create some examples to introduce different levels of normalization step by step.

Guidance for assessments and evaluations:

- Provide students with a printout with zero normalized tables and ask them to find the shortcomings and normalize up to third normalized form step by step.

Quality inputs:

- Printouts with zero normalized tables, presentations

Competency 9: Develops algorithms to solve problems and uses python programming language to encode algorithms

Competency Level 9.1: Uses problem-solving process
Competency Level 9.2: Explores the top down and stepwise refinement methodologies in solving problems

Time: 06 periods

Learning Outcomes:

- Describes the steps of problem solving process
- Implements the solution Briefly describes Candidate key, primary key, alternate key and foreign key
- Uses stepwise refinement methodology to solve problems
- Draws structures charts to illustrate a solution for a system

Contents:

- Understanding the problem
- Defining the problem and boundaries
- Planning solution
- Implementation
- Modularization
- Top down design and stepwise refinement
- Structure charts

Concepts and terms to be highlighted:

- Appreciate the process of problem solving, using critical thinking and logical process
- Design a system by breaking the system into small subsystem and connecting them logically
- Structure charts
- Illustrate how structure charts can be used to show the breakdown of a system to its lowest manageable levels

Guidance for lesson plans:

- Divide the students into groups
- Provide simple problems to each group and ask them to discuss the problem solving process and present it
- Discuss the systematic process of problem solving
- Ask each group to identify a problem to solve
- Ask the group to break the problem into small sub parts (subsystems)
- Discuss the top down and stepwise refinement methodology in problem solving
- Introduce Structure Charts

Guidance for assessments and evaluations:

- Provide a problem and ask the students to write down the steps to solve it

- Ask the student to draw the Structure Chart for the problem

Quality inputs:

- Printout with problems, presentations

Competency Level 9.3: Uses algorithmic approach to solve problems

Time: 06 periods

Learning Outcomes:

- Briefly describes algorithms
- Identifies the standard symbols used to draw flow charts
- Draws flow charts to illustrate solutions to a given problem
- Writes pseudo codes to illustrate solutions to a given problem
- Uses hand traces to verify the solutions

Contents:

- Algorithms
  - Flow charts
  - Pseudo codes
  - Hand traces

Concepts and terms to be highlighted:

- Algorithm
- Algorithm is the step by step process to be followed in solving a problem
- Sequence
- Selection
- Repetition
- Flow chart
- Flow chart is a diagram used to represent an algorithm
- Pseudo code
- Pseudo code is a high level description used to represent an algorithm

Guidance for lesson plans:

- Provide a problem and ask the students to write down the solution in stepwise
- Discuss their stepwise solution and introduce the term algorithm and its representations (flow chart and pseudo code)
- Provide some situations of sequence, selection, repetition and discuss each of them
- Provide some sample problems and discuss how to show their solutions in flow charts and pseudo code
- Provide some problems and ask the students to create algorithms in both flowcharts and pseudo code and present it

Guidance for assessments and evaluations:

- Provide problems involving sequence only, selection, iteration selection and iteration and ask the students to create algorithms in both flow chart and pseudo code
- Provide flow charts and ask the students to find the output of the problem